

Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
Department of Geosciences, Boise State University
Env. Research Building (ERB) 1104, 3116 (labs), 3155 (office)
1910 University Drive, Boise, ID 83725

(208) 426-1416 (office)
(303) 859-3106 (cell)
hpmarshall@boisestate.edu

Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington, Thesis: *Wet snow densification*
- 1994 Nathan Hale High School, Seattle, Washington, GPA: 3.9/4.0

Professional Experience

- 08/2023-present Professor of Geophysics, Co-Director of CryoGARS, Department of Geosciences
- 07/2015-present Co-founder and CEO, CryoToolbox, LLC
- 08/2013-08/2023 Associate Professor, Dept. of Geosciences, Director of CryoGARS, Boise State University
- 06/2021-present Leadership team member, NASA SnowEx Mission
- 10/2021-present Avalanche Instructor and Senior Advisor, Avalanche Science LLC
- 06/2018-08/2021 Project Scientist, NASA SnowEx Mission (20+ organizations, 100+ participants)
- 08/2016-08/2022 Senior Project Staff Engineer, ATA Aerospace, LLC
- 07/2007-03/2020 Geophysical Engineer (Expert Consultant), U.S. Army Corps of Engineers
Cold Regions Research and Engineering Lab (CRREL), Hanover, New Hampshire
- 08/2008-07/2013 Assistant Professor, Department of Geosciences and Center for Geophysical
Investigation of the Shallow Subsurface (CGISS), Boise State University
- 01/2005-08/2011 Chair of Research Committee, American Avalanche Association
- 04/2006-08/2008 Research Associate, Inst. of Arctic and Alpine Research (INSTAAR)
University of Colorado at Boulder
- 12/2004-04/2006 Research Geophysical Engineer (GS-11), Student Career Experiences Program (SCEP)
U.S. Army Corps of Engineers CRREL, Hanover, New Hampshire & Fairbanks, Alaska
- 01/2004-07/2004 Invited visiting scientist, Swiss Federal Institute for Snow and
Avalanche Research (SLF), Davos, Switzerland
- 07/2000-12/2005 Research Assistant, INSTAAR, University of Colorado at Boulder
- 11/2002-12/2003 Avalanche Instructor, Alpine World Ascents, Boulder, Colorado
- 06/1995-07/2000 Undergraduate Research Assistant, Glaciology Group, University of Washington
- 08/1999-03/2000 Volunteer high school teacher, Old Crow, Yukon Terr., Canada
- Summers 1993-94 Roofer, Seattle, WA

Research Interests

Snow science and glaciology, using geophysics, engineering, and remote sensing
Spatial variability of snow, snow and ice mechanics, snow slope stability, meltwater dynamics
Computational statistics, geostatistics, compressed sensing, instrumentation development

Honors and Awards

2020	BSU Faculty Excellence Award
2019	NASA Instrument Incubator Program Award, NASA, 1st in Idaho history
2019	2018 WRR Editors' Choice Award: Hedrick et al., 2018
2018	NASA Robert H. Goddard Team Award: SnowEx Year 1 Organizing Team
2012	Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
2010-13	New Investigator Program Award, NASA, 1st in Idaho history
2010	Young Scientist Award, Cryosphere Section, American Geophysical Union
2009	Top Reviewer Award, Cold Regions Science and Technology (CRST)
2008	U.S. Young Scientist Award, International Union of Radio Science (URSI)
2002-05	N.A.S.A. Earth Systems Science Graduate Fellowship
2003	Backcountry Access Avalanche Education Support Grant
2003	Arctic Research Consortium of the U.S. (ARCUS) Award for Arctic Research Excellence, Hon. Mention
2002	University of Colorado Graduate School Student Travel Grant
2001-02	N.S.F. K-12 Engineering Outreach Fellowship
2002	Beverly Sears Graduate Research Grant (University of Colorado)
2001	American Avalanche Association Graduate Research Grant
2000	Li Tze Fong Memorial Fellowship, University of British Columbia (declined)
1999	University of Washington Bonderman Honors Travel Fellowship
1999	Volunteer of the Year, Arctic Canada, Frontiers Foundation
1997-99	University of Washington Mary Gates Undergraduate Research Grant
1999	Featured speaker, University of Washington Undergraduate Research Symposium
1995-98	N.A.S.A. Space Grant Undergraduate Research Program
1999,2001	N.S.F. Graduate Fellowship Program - Honorable Mention
1994-95,97-99	Dean's List, University of Washington
1994-95	U.W. Undergraduate Scholar Award (1994-95)
1994	Washington Athletic Club Student Athlete of the Year, Nathan Hale High School

Student Awards

2023-2026	Zach Keskinen, Ph.D. Student <i>Future Investigator in NASA Earth and Space Science and Technology (FINESST)</i>
2018-2020	Joel Gongora, Ph.D. student <i>Future Investigator in NASA Earth and Space Science and Technology (FINESST)</i>
2018	Tate Meehan, Ph.D. student <i>Idaho Space Grant Consortium, NASA EPSCoR graduate fellowship</i>
2012-2014	Scott Havens, Ph.D. student <i>NASA Earth Systems Science Fellowship</i>
2012	Andrew Hedrick, M.S. student <i>American Avalanche Association Graduate Student Grant</i>
2012	Scott Havens, Ph.D. student <i>1st place poster NSF/NASA EPSCoR annual meeting</i>
2012	Kris Burch, Lloyd Lowe II, Kolton Drake, Undergrads, <i>Best Senior Project, BSU College of Engineering</i>
2003	Adam Maerz, High School student <i>Colorado State Science Fair Finalist</i>

Refereed Publications [peer-reviewed publications=106, h-index=36, i10-index=83, citations=4592]

- 2025 Alabi, I., **Marshall, H.P.**, Mead, J., and Trujillo, E. (2025). Advancing terrestrial snow depth monitoring with machine learning and l-band insar data: a case study using nasa's snowex 2017 data. *Frontiers in Remote Sensing*, 5(1481848):doi: 10.3389/frsen.2024.1481848
- Brangers, I., De Lannoy, G., **Marshall, H.P.**, Dunmire, D., Bonnell, R., Cox, B., Cappelle, J., Brad Baxter, W., and Lievens, H. (2025). C-band radar measurements in a snow-covered boreal forest environment. *IEEE Geoscience and Remote Sensing Letters*, 22:1–5
- Boueshagh, M., Ramage, J. M., Brodzik, M. J., Long, D. G., Hardman, M., and **Marshall, H.P.** (2025). Revealing causes of a surprising correlation: snow water equivalent and spatial statistics from calibrated enhanced-resolution brightness temperatures (cetb) using interpretable machine learning and shap analysis. *Frontiers in Remote Sensing*, 6(1554084):doi: 10.3389/frsen.2025.1554084
- Wilder, B. A., Enterkine, J., Hoppinen, Z., Adebisi, N., **Marshall, H.P.**, O'Neel, S., Van Der Weide, T., Kinoshita, A. M., and Glenn, N. F. (2025). Modeling snow optical properties from single wavelength airborne lidar in steep forested terrain. *Frontiers in Earth Science*, 13(1487776):doi: 10.3389/feart.2025.1487776
- Tarricone, J., Palomaki, R., Rittger, K., Nolin, A., **Marshall, H.P.**, and Vuyovich, C. (2025). Investigating the impact of optical snow cover data on l-band insar snow water equivalent retrievals. *Journal of Remote Sensing*, 5:1–18, DOI: 10.34133/remotesensing.0682
- Stuefer, S., Hale, K., May, L., Mason, M., Vuyovich, C., **Marshall, H.P.**, Vas, D., and Elder, K. (2025). Snow depth measurements from arctic tundra and boreal forest collected during nasa snowex alaska campaign. *Scientific Data*, 12(919):1–13, DOI: 10.1038/s41597-025-05170-x
- Aberle, R., Enderlin, E., O'Neel, S., Florentine, C., Sass, L., Dickson, A., **Marshall, H.P.**, and Flores, A. (2025). Automated snow cover detection on mountain glaciers using spaceborne imagery and machine learning. *The Cryosphere*, 19(4):1675–1693, DOI: 10.5194/tc-19-1675-2025
- 2024 Hoppinen, Z., Palomaki, R. T., Brencher, G., Dunmire, D., Gaglianon, E., Marziliano, A., Tarricone, J., and **Marshall, H.P.** (2024a). Evaluating snow depth retrievals from sentinel-1 volume scattering over nasa snowex sites. *The Cryosphere*, 18(11):5407–5430, <https://doi.org/10.5194/tc-18-5407-2024>
- Bonnell, R., Elder, K., McGrath, D., **Marshall, H.P.**, Starr, B., Adebisi, N., Palomaki, R. T., and Hoppinen, Z. (2024a). L-band insar snow water equivalent retrieval uncertainty increases with forest cover fraction. *Geophysical Research Letters*, 51(e2024GL111708):<https://doi.org/10.1029/2024GL111708>
- Meehan, T. G., Hojatimalekshah, A., **Marshall, H.P.**, Deeb, E. J., O'Neel, S., McGrath, D., Webb, R. W., Bonnell, R., Raleigh, M. S., Hiemstra, C., and Elder, K. (2024). Spatially distributed snow depth, bulk density, and snow water equivalent from ground-based and airborne sensor integration at Grand Mesa, Colorado, USA. *The Cryosphere*, 18(7):3253–3276, doi.org/10.5194/tc-18-3253-2024

- Oveisgharan, S., Zinke, R., Hoppinen, Z., and **Marshall, H.P.** (2024). Snow Water Equivalent Retrieval Over Idaho, Part A: Using Sentinel-1 Repeat-Pass Interferometry. *The Cryosphere*, 18:559–574, doi.org/10.5194/tc-18-559-2024
- Hoppinen, Z. M., Oveisgharan, S., **Marshall, H.P.**, Mower, R., Elder, K., and Vuyovich, C. (2024b). Snow Water Equivalent Retrieval Over Idaho, Part B: Using L-band UAVSAR Repeat-Pass Interferometry. *The Cryosphere*, 18:575–592, doi.org/10.5194/tc-18-575-2024
- Bonnell, R., McGrath, D., Tarricone, J., **Marshall, H.P.**, Bump, E., Duncan, C., Kampf, S., Lou, Y., Olsen-Mikitowicz, A., Sears, M., Williams, K., Zeller, L., and Zheng, Y. (2024b). Evaluating L-band InSAR snow water equivalent retrievals with repeat ground-penetrating radar and terrestrial lidar surveys in northern colorado. *The Cryosphere*, 18(7):3765–3785, doi.org/10.5194/tc-18-3765-2024
- McCaslin, J. C., Mikesell, T. D., and **Marshall, H.P.** (2024). A laser ultrasound system to non-invasively measure compression waves in granular ice mixes. *Cold Regions Science and Technology*, 221:104157, doi.org/10.1016/j.coldregions.2024.104157
- Brangers, I., **Marshall, H.P.**, De Lannoy, G., Dunmire, D., Matzler, C., and Lievens, H. (2024b). Tower-based C-band radar measurements of an alpine snowpack. *The Cryosphere*, 18:3177–3193, https://doi.org/10.5194/tc-18-3177-2024
- Wilder, B. A., Lee, C. M., Chlus, A., **Marshall, H.P.**, Brandt, J., Kinoshita, A. M., Enterkine, J., Van Der Weide, T., and Glenn, N. F. (2024). Computationally efficient retrieval of snow surface properties from spaceborne imaging spectroscopy measurements through dimensionality reduction using k-means spectral clustering. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 17:8594–8605
- McDowell, I. E., Keegan, K. M., Skiles, S. M., Donahue, C. P., Osterberg, E. C., Hawley, R. L., and **Marshall, H.P.** (2024). A cold laboratory hyperspectral imaging system to map grain size and ice layer distributions in firn cores. *The Cryosphere*, 18(4):1925–1946, doi.org/10.5194/tc-18-1925-2024
- Rudisill, W., Flores, A., **Marshall, H.P.**, Siirila-Woodburn, E., Feldman, D., Rhoades, A., Xu, Z., and Morales, A. (2024). Cold-season precipitation sensitivity to microphysical parameterizations: Hydrologic evaluations leveraging snow lidar datasets. *Journal of Hydrometeorology*, 25(1):143–160
- Kim, Y., Bharathan, A., Romero, C., Chahat, N., Gupta, G., Chattopadhyay, G., **Marshall, H.P.**, Lin, N., Gier, J., and Tang, A. (2024). Close-in dynamic range considerations in up-converted and frequency-multiplied frequency-modulated continuous-wave radars at Ku-Band. *IEEE Microwave Magazine*, 25(9):22–28, doi: 10.1109/MMM.2024.3412002
- Tang, A., Chahat, N., Kim, Y., Bharathan, A., Virbila, G., **Marshall, H.P.**, Van Der Weide, T., Gupta, G., Anand, R., Chattopadhyay, G., and Chang, M.-C. F. (2024). A UAV Based CMOS Ku-Band Metasurface FMCW Radar System for Low-Altitude Snowpack Sensing. *IEEE Journal of Microwaves*, 4(1):43–55, doi: 10.1109/JMW.2023.3327188

2023

Siqueira, P., Tarres, M. C., Adam, M., Sutherland, E., Maloyan, J., Seaver, T., Tessier, R., Tsang, L., Borah, F., **Marshall, H.P.**, Deeb, E., and Farquharson, G. (2024). SNOWWI: A three-frequency InSAR for snow science applications. *Proceedings of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 1842–1845, DOI: 10.1109/IGARSS53475.2024.10642002

Tarres, M. C., Siqueira, P., Adam, J. M., Sutherland, E., Maloyan, J., Seaver, T., Tessier, R., Tsang, L., Borah, F., **Marshall, H.P.**, Deeb, E., and Farquharson, G. (2024). First results from a dual Ku-and C-band airborne SAR for snowpack measurements. *Proceedings of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 2460–2463, DOI: 10.1109/IGARSS53475.2024.10640580

Hoppinen, Z., Johnson, J. B., Capelli, A., Anderson, J. H., and **Marshall, H.P.** (2023). Snow attenuation of infrasound signals and wind noise. *Seismological Research Letters*, 94(6):2801–2812, doi.org/10.1785/0220230048

Aberle, R., Enderlin, E. M., **Marshall, H.P.**, Kopera, M., and Meehan, T. G. (2023). Assessing controls on ice dynamics at crane glacier, antarctic peninsula, using a numerical ice flow model. *Journal of Glaciology*, 69(277):1109–1124

Bonnell, R., McGrath, D., Hedrick, A. R., Trujillo, E., Meehan, T. G., Williams, K., **Marshall, H.P.**, Sexstone, G., Fulton, J., Ronayne, M. J., Fassnacht, S. R., Webb, R. W., and Hale, K. E. (2023). Snowpack relative permittivity and density derived from near-coincident lidar and ground-penetrating radar. *Hydrological Processes*, 37(10):e14996

McDowell, I. E., Keegan, K. M., Wever, N., Osterberg, E. C., Hawley, R. L., and **Marshall, H.P.** (2023). Firn core evidence of two-way feedback mechanisms between meltwater infiltration and firn microstructure from the western percolation zone of the greenland ice sheet. *Journal of Geophysical Research: Earth Surface*, 128(2):e2022JF006752

Hojatimalekshah, A., Gongora, J., Enterkine, J., Glenn, N. F., Caughlin, T., **Marshall, H.P.**, and Hiemstra, C. A. (2023). Lidar and deep learning reveal forest structural controls on snowpack. *Frontiers in Ecology and the Environment*, 21(1):49–54

Tarricone, J., Webb, R., **Marshall, H.P.**, Nolin, A., and Meyer, F. (2023). Estimating snow accumulation and ablation with L-band InSAR. *The Cryosphere*, 17(5):1997–2019, doi.org/10.5194/tc-17-1997-2023

Marshall, H.P., Storms, S., Deeb, E., Forster, R., Vuyovich, C., Elder, K., Durand, M., and Hiemstra, C. (2023). Using phase-delay approaches to estimate snow properties: A comparison of airborne l-band insar and ground-based 6-18 ghz fmcw radar observations during the nasa snowex 2020 grand mesa campaign. *IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium*, doi: 10.1109/IGARSS52108.2023.10281420:848–851

Wilder, B. A., Glenn, N. F., Lee, C. M., **Marshall, H.P.**, Brandt, J., Kinoshita, A. M., Van Der Weide, T., and Enterkine, J. (2023). Global optical snow properties via high-speed algorithm with k-means clustering (goshawk). *IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium*, doi:10.1109/IGARSS52108.2023.10283419:118–120

- Oveisgharan, S., Zinke, R., Keskinen, Z., and **Marshall, H.P.** (2023). Estimating snow water equivalent using sentinel-1 repeat-pass interferometry. *IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium*, doi: 10.1109/IGARSS52108.2023.10282893:856–859
- Meehan, T. G., Deeb, E. J., **Marshall, H.P.**, and O’Neel, S. (2023). Assessing the representation of wind, terrain, and vegetation effects on snow density distributed by learned regression modeling. *IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium*, doi: 10.1109/IGARSS52108.2023.10281806:28–31
- 2022 Lievens, H., Brangers, I., **Marshall, H.P.**, Jonas, T., Olefs, M., and DeLannoy, G. (2022). Sentinel-1 snow depth retrieval at sub-kilometer resolution over the European Alps. *The Cryosphere*, 16(1):159–177, doi.org/10.5194/tc-16-159-2022
- McGrath, D., Bonnell, R., Zeller, L., Olsen-Mikitowicz, A., Bump, E., Webb, R., and **Marshall, H.P.** (2022). A time-series of snow density and snow water equivalent observations derived from the integration of GPR and UAV SfM observations. *Frontiers in Remote Sensing*, 3:886747, doi:10.3389/frsen.2022.886747
- Liu, J., Enderlin, E. M., **Marshall, H.P.**, and Khalil, A. (2022). Synchronous retreat of south-east greenland’s peripheral glaciers. *Geophysical Research Letters*, 49(13):e2022GL097756, <https://doi.org/10.1029/2022GL097756>
- Lund, J., Forster, R., Deeb, E., Liston, G., Skiles, M., and **Marshall, H.P.** (2022). Interpreting Sentinel-1 SAR backscatter signals of snowpack surface melt/freeze, warming, and ripening, through field measurements and physically-based snowmodel. *Remote Sensing*, 14:4002, <https://doi.org/10.3390/rs14164002>
- Tsang, L., Durand, M., Derksen, C., Barros, A. P., Kang, D.-H., Lievens, H., **Marshall, H.P.**, Zhu, J., Johnson, J., King, J., Lemmetyinen, J., Sandells, M., Rutter, N., Siqueira, P., Nolin, A., Osmanoglu, B., Vuyovich, C., Kim, E. J., Taylor, D., Merkouriadi, I., Brucker, L., Navari, M., Dumont, M., Kelly, R., Kim, R. S., Liao, T.-H., and Xu, X. (2022). Review article: Global monitoring of snow water equivalent using high frequency radar remote sensing. *The Cryosphere*, 16(9):3531–3573
- Lemmetyinen, J., Cohen, J., Kontu, A., Hannula, J. V. H., Merkouriadi, I., Scheiblauer, S., Rott, H., Nagler, T., Ripper, E., Elder, K., **Marshall, H.P.**, Fromm, R., Adams, M., Derksen, C., King, J., Meta, A., Coccia, A., Rutter, N., Sandells, M., Macelloni, G., Santi, E., Leduc-Leballeur, M., Essery, R., Menard, C., and Kern, M. (2022). Airborne SnowSAR data at X- and Ku-bands over boreal forest, alpine and tundra snow cover. *Earth Syst. Sci. Data*, 14(9):3915–3945, doi.org/10.5194/essd-14-3915-2022
- Brangers, I., **Marshall, H.P.**, De Lannoy, G., and Lievens, H. (2022). Tower based c-band radar observations of the snowpack. In *IGARSS 2022 - 2022 IEEE International Geoscience and Remote Sensing Symposium*, pages 4559–4562, DOI:10.1109/IGARSS46834.2022.9884416
- Adebisi, N. and **Marshall, H.P.** (2022). Snow depth retrieval from L-band data based on repeat pass InSAR techniques. In *IGARSS 2022 - 2022 IEEE International Geoscience and Remote Sensing Symposium*, pages 4248–4251, doi: 10.1109/IGARSS46834.2022.9884723

2021

Enderlin, E., Elkin, C., Gendreau, M., **Marshall, H.P.**, O’Neel, S., McNeil, C., Sass, L., and Florentine, C. (2022). Uncertainty of icesat-2 atl06- and atl08-derived snow depths for glacierized and vegetated mountain regions. *Remote Sens. Environ.*, 283:113307

Kraft, M., McNamara, J., **Marshall, H.P.**, and Glenn, N. (2022). Forest impacts on snow accumulation and melt in a semi-arid mountain environment. *Front. Water*, 4(1004123, doi:10.3389/frwa.2022.1004123)

Marshall, H.P., Deeb, E., Forster, R., Vuyovich, C., Elder, K., Hiemstra, C., and Lund, J. (2021). L-band InSAR depth retrieval during the NASA SnowEx 2020 Campaign: Grand Mesa, Colorado. *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 625–627, DOI:10.1109/IGARSS47720.2021.9553852

Lewis, G., Osterberg, E., Hawley, B., **Marshall, H.P.**, Meehan, T., Graeter, K. A., McCarthy, F., Overly, T., Thundercloud, Z., Ferris, D., Koffman, B., and Dibb, J. (2021). Atmospheric blocking drives recent albedo change across the western Greenland ice sheet percolation zone. *Geophysical Research Letters*, 48(10):e2021GL092814

Johnson, J., Anderson, J., **Marshall, H.P.**, Havens, S., and Watson, L. (2021a). Snow avalanche detection and source constraints made using multiple infrasound sensor arrays. *Journal of Geophysical Research-Earth Surface*, 126(3):e2020JF005741

Liu, J., Enderlin, E. M., **Marshall, H.P.**, and Khalil, A. (2021). Automated detection of marine glacier calving fronts using the 2d wavelet transform modulus maxima (wtmm) segmentation method. *IEEE Transactions On Geoscience and Remote Sensing*, 59(11):9047–9056, doi:10.1109/TGRS.2021.3053235

Meehan, T. G., **Marshall, H.P.**, Bradford, J. H., Hawley, R. L., Overly, T. B., Lewis, G., Graeter, K., Osterburg, E., and McCarthy, F. (2021). Historical surface mass balance reconstruction 1984-2017 from greentracs multi-offset ground-penetrating radar. *Journal of Glaciology*, 67(262):219–228, doi:10.1017/jog.2020.91

Hojatimalekshah, A., Uhlmann, Z., Glenn, N. F., Hiemstra, C. A., Tennant, C. J., Graham, J. D., Spaete, L., Gelvin, A., **Marshall, H.P.**, McNamara, J., and Enterkine, J. (2021). Tree canopy and snow depth relationships at fine scales with terrestrial laser scanning. *The Cryosphere*, 15(5):2187–2209, doi:10.5194/tc-15-2187-2021

Kim, R. S., Kumar, S., Vuyovich, C., Houser, P., Lundquist, J., Mudryk, L., Durand, M., Barros, A., Kim, E. J., Forman, B. A., Gutmann, E. D., Wrzesien, M. L., Garnaud, C., Sandells, M., **Marshall, H.P.**, Cristea, N., Pflug, J. M., Johnston, J., Cao, Y., Mocko, D., and Wang, S. (2021). Snow Ensemble Uncertainty Project (SEUP): Quantification of snow water equivalent uncertainty across north america via ensemble land surface modeling. *The Cryosphere*, 15:771–791, doi:10.5194/tc-15-771-2021

Webb, R. W., Marziliano, A., McGrath, D., Bonnell, R., Meehan, T. G., Vuyovich, C., and **Marshall, H.P.** (2021). In situ determination of dry and wet snow permittivity: Improving equations for low frequency radar applications. *Remote Sensing*, 13:4617, DOI:10.3390/rs13224617

- Bonnell, R., McGrath, D., Williams, K., Webb, R., Fassnacht, S. R., and **Marshall, H.P.** (2021b). Spatiotemporal variations in liquid water content in a seasonal snowpack: Implications for radar remote sensing. *Remote Sensing*, 13:4223, DOI:10.3390/rs13214223
- Lievens, Hans and Brangers, Isis and **Marshall, H.P.** and Jonas, Tobias and Olefs, Marc and De Lannoy, Gabriëlle (2021). Observing snow depth at sub-kilometer resolution over the european alps from sentinel-1. *2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS*, pages 618–621, doi:10.1109/IGARSS47720.2021.9553420
- 2020 Hedrick, A. R., Marks, D., **Marshall, H.P.**, McNamara, J., Havens, S., Trujillo, E., Sandusky, M., Robertson, M., Johnson, M., Bormann, K. J., and Painter, T. H. (2020). From drought to flood: A water balance analysis of the tuolumne river basin during extreme conditions (2015 – 2017). *Hydrol. Process.*, 34(11):2560–2574
- Webb, R., Raleigh, M., McGrath, D., Molotch, N., Elder, K., Hiemstra, C., Brucker, L., and **Marshall, H.P.** (2020). Within-stand boundary effects of snow water equivalent distribution in forested areas. *Water Resources Research*, 56:e2019WR024905
- Lund, J., Forster, R. R., Rupper, S. B., **Marshall, H.P.**, Deeb, E. J., and Hashmi, M. (2020). Mapping snowmelt progression in the upper indus basin with synthetic aperture radar. *Frontiers in Earth Science*, 7:318, doi:10.3389/feart.2019.00318
- 2019 McGrath, D., Webb, R., Shean, D., Bonnell, R., **Marshall, H.P.**, Painter, T., Molotch, N., Elder, K., Hiemstra, C., and Brucker, L. (2019). Spatially extensive ground-penetrating radar snow depth observations during NASA’s 2017 SnowEx Campaign: Comparison to in situ, airborne, and satellite observations. *Water Resources Research*, 55(11):10026–10036, doi:10.1029/2019WR024907
- Lievens, H., Demuzere, M., **Marshall, H.P.**, Reichle, R. H., Brucker, L., Brangers, I., de Rosnay, P., Dumont, M., Giroto, M., Immerzeel, W. W., Jonas, T., Kim, E. J., Koch, I., Marty, C., Saloranta, T., Schöber, J., and Lannoy, G. J. D. (2019). Snow depth variability in the northern hemisphere mountains observed from space. *Nature Communications*, 10(1):1–12
- Lewis, G., Osterberg, E., Hawley, R., **Marshall, H.P.**, Meehan, T., Graeter, K., McCarthy, F., Overly, T., Thundercloud, Z., and Ferris, D. (2019). Recent precipitation decrease across the western greenland ice sheet percolation zone. *The Cryosphere*, 13:2797–2815, doi:10.5194/tc-13-2797-2019
- 2018 Graeter, K. A., Osterberg, E. C., Ferris, D. G., Hawley, R. L., **Marshall, H.P.**, Lewis, G., Meehan, T., McCarthy, F., and Birkel, S. D. (2018). Ice core records of West Greenland surface melt and climate forcing. *Geophysical Research Letters*, 45(7):3164–3172, doi:10.1002/2017GL076641
- Hedrick, A., Marks, D., Kormos, P., **Marshall, H.P.**, Havens, S., Bormann, K., and Painter, T. H. (2018). Direct insertion of NASA Airborne Snow Observatory-derived snow depth time-series into the iSnobal energy balance snow model. *Water Resources Research*, 54(10):8045–8063, doi.org/10.1029/2018WR023190

McGrath, D., Sass, L., O’Neel, S., McNeil, C., Candela, S. G., Baker, E. H., and **Marshall, H.P.** (2018). Interannual snow accumulation variability on glaciers derived from repeat, spatially extensive ground-penetrating radar surveys. *The Cryosphere*, 12:3617–3633, doi.org/10.5194/tc-12-3617-2018

Hagenmuller, P., van Herwijnen, A., Pielmeier, C., and **Marshall, H.P.** (2018). Evaluation of the snow penetrometer Avatech SP2. *Cold Regions Science and Technology*, 149:83–94

McNamara, J. P., Benner, S. G., Poulos, M. J., Pierce, J. L., Chandler, D. G., Kormos, P. R., **Marshall, H.P.**, Flores, A. N., Seyfried, M., Glenn, N. F., and Aishlin, P. (2018). Form and function relationships revealed by long-term research in a semiarid mountain catchment. *WIREs Water*, 5(2):e1267, doi:10.1002/wat2.1267

Brucker, L., Hiemstra, C., **Marshall, H.P.**, Elder, K., De Roo, R., Mousavi, M., Bliven, F., Peterson, W., Deems, J., Gadomski, P., Gelvin, A., Spaete, L., Barnhart, T., Brandt, T., Burkhart, J., Crawford, C., Datta, T., Erikstrod, H., Glenn, N., Hale, K., Holben, B., Houser, P., Jennings, K., Kelly, R., Kraft, J., Langlois, A., McGrath, D., Merriman, C., Molotch, N., Nolin, A., Polashenski, C., Raleigh, M., Rittger, K., Rodriguez, C., Roy, A., Skiles, M., Small, E., Tedesco, M., Tennant, C., Thompson, A., Uhlmann, Z., Webb, R., and Wingo, M. (2018b). Nasa SnowEx’17 in situ Measurements and Ground-Based Remote Sensing. *IGARSS 2018 - 2018 IEEE International Geoscience and Remote Sensing Symposium*, pages 6266–6268, doi:10.1109/IGARSS.2018.8517777

Kim, Y., Reck, T. J., Alonso-delPino, M., Painter, T. H., **Marshall, H.P.**, Bair, E. H., Dozier, J., Chattopadhyay, G., Liou, K.-N., Chang, M.-C. F., and Tang, A. (2018). A Ku-band CMOS FMCW radar transceiver for snowpack remote sensing. *IEEE Transactions On Microwave Theory and Techniques*, 66(5):2480–2494

2017 Mikesell, T., van Wijk, K., Otheim, L., **Marshall, H.P.**, and Kurbatov, A. (2017). Laser ultrasound observations of mechanical property variations in ice cores. *Geosciences*, 7(3):47

Lewis, G., Osterberg, E., Hawley, R., Whitmore, B., **Marshall, H.P.**, and Box, J. (2017). Regional Greenland accumulation variability from Operation IceBridge airborne accumulation radar. *The Cryosphere*, 11(2):773

Deeb, E. J., **Marshall, H.P.**, Forster, R. R., Jones, C. E., Hiemstra, C. A., and Siqueira, P. R. (2017). Supporting NASA SnowEx remote sensing strategies and requirements for L-band interferometric snow depth and snow water equivalent estimation. *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 1395–1396, doi:10.1109/IGARSS.2017.8127224

Kim, E., Gatebe, C., Hall, D., Newlin, J., Misakonis, A., Elder, K., **Marshall, H.P.**, Hiemstra, C., Brucker, L., De Marco, E., Crawford, C., Kang, D. H., and Entin, J. (2017). NASA’s SnowEx campaign: Observing seasonal snow in a forested environment. *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 1388–1390, doi:10.1109/IGARSS.2017.8127222

- Pegau, W. S., Garron, J., Zabilansky, L., Bassett, C., Bello, J., Bradford, J., Carns, R., Courville, Z., Eicken, H., Elder, B., Eriksen, P., Lavery, A., Light, B., Maksym, T., **Marshall, H.P.**, Oggier, M., Perovich, D., Pacwiardowski, P., Singh, H., Tang, D., Wiggins, C., and Wilkinson, J. (2017). Detection of oil in and under ice. *International Oil Spill Conference Proceedings*, 2017(1):1857–1876
- Brucker, L., Hiemstra, C., **Marshall, H.P.**, Elder, K., De Roo, R., Mousavi, M., Bliven, F., Peterson, W., Deems, J., Gadowski, P., Gelvin, A., Spaete, L., Barnhart, T., Brandt, T., Burkhart, J., Crawford, C., Datta, T., Erikstrod, H., Glenn, N., Hale, K., Holben, B., Houser, P., Jennings, K., Kelly, R., Kraft, J., Langlois, A., McGrath, D., Merriman, C., Molotch, N., Nolin, A., Polashenski, C., Raleigh, M., Rittger, K., Rodriguez, C., Roy, A., Skiles, M., Small, E., Tedesco, M., Tennant, C., Thompson, A., Tian, L., Uhlmann, Z., Webb, R., and Wingo, M. (2017). A first overview of SnowEx ground-based remote sensing activities during the winter 2016–2017. *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 1391–1394, doi:10.1109/IGARSS.2017.8127223
- 2016 Evans, S., Flores, A., Heilig, A., Kohn, M., **Marshall, H.P.**, and McNamara, J. (2016). Isotopic evidence for lateral flow and diffusive transport, but not sublimation, in a sloped seasonal snowpack, Idaho, U.S.A. *Geophysical Research Letters*, 43(7):3298–3306
- Rutter, N., **Marshall, H.P.**, Tape, K., Essery, R., and King, J. (2016). Impact of spatial averaging on radar reflectivity at internal snowpack layer boundaries. *Journal of Glaciology*, 62(236):1065–1074
- Bradford, J. H., Babcock, E. L., **Marshall, H.P.**, and Dickins, D. F. (2016). Targeted reflection-waveform inversion of experimental ground-penetrating radar data for quantification of oil spills under sea ice. *Geophysics*, 81(1):WA59–WA70
- 2015 Heilig, A., Mitterer, C., Schmid, L., Wever, N., Schweizer, J., **Marshall, H.P.**, and Eisen, O. (2015). Seasonal and diurnal cycles of liquid water in snow-measurements and modeling. *Journal of Geophysical Research-Earth Surface*, 120(10):2139–2154
- Kormos, P. R., McNamara, J. P., Seyfried, M. S., **Marshall, H.P.**, Marks, D., and Flores, A. N. (2015). Bedrock infiltration estimates from a catchment water storage-based modeling approach in the rain snow transition zone. *Journal of Hydrology*, 525:231–248
- Hedrick, A., **Marshall, H.P.**, Winstral, A., Elder, K., Yueh, S., and Cline, D. (2015). Independent evaluation of the SNODAS snow depth product using regional-scale LiDAR-derived measurements. *The Cryosphere*, 9(1):13–23
- 2014 Kormos, P. R., Marks, D., McNamara, J. P., **Marshall, H.P.**, Winstral, A., and Flores, A. N. (2014a). Snow distribution, melt and surface water inputs to the soil in the mountain rain-snow transition zone. *Journal of Hydrology*, 519:190–204
- Walters, R. D., Watson, K. A., **Marshall, H.P.**, McNamara, J. P., and Flores, A. N. (2014). A physiographic approach to downscaling fractional snow cover data in mountainous regions. *Remote Sensing of Environment*, 152:413–425
- Havens, S., **Marshall, H.P.**, Johnson, J. B., and Nicholson, B. (2014a). Calculating the velocity of a fast-moving snow avalanche using an infrasound array. *Geophysical Research Letters*, 41(17):6191–6198

- Anderson, B. T., McNamara, J. P., **Marshall, H.P.**, and Flores, A. N. (2014). Insights into the physical processes controlling correlations between snow distribution and terrain properties. *Water Resources Research*, 50(6):4545–4563
- Tinkham, W. T., Smith, A. M. S., **Marshall, H.P.**, Link, T. E., Falkowski, M. J., and Winstral, A. H. (2014). Quantifying spatial distribution of snow depth errors from lidar using random forest. *Remote Sensing of Environment*, 141:105–115
- McCreight, J. L., Slater, A. G., **Marshall, H.P.**, and Rajagopalan, B. (2014). Inference and uncertainty of snow depth spatial distribution at the kilometre scale in the colorado rocky mountains: the effects of sample size, random sampling, predictor quality, and validation procedures. *Hydrological Processes*, 28(3):933–957
- Kormos, P. R., Marks, D., Williams, C. J., **Marshall, H.P.**, Aishlin, P., Chandler, D. G., and McNamara, J. P. (2014b). Soil, snow, weather, and sub-surface storage data from a mountain catchment in the rain-snow transition zone. *Earth System Science Data*, 6(1):165–173
- 2013 Havens, S., **Marshall, H.P.**, Pielmeier, C., and Elder, K. (2013). Automatic grain type classification of snow micro penetrometer signals with random forests. *IEEE Transactions On Geoscience and Remote Sensing*, 51(6):3328–3335
- Eiriksson, D., Whitson, M., Luce, C. H., **Marshall, H.P.**, Bradford, J., Benner, S. G., Black, T., Hetrick, H., and McNamara, J. P. (2013). An evaluation of the hydrologic relevance of lateral flow in snow at hillslope and catchment scales. *Hydrological Processes*, 27(5):640–654
- Tinkham, W. T., Hoffman, C. M., Falkowski, M. J., Smith, A., **Marshall, H.P.**, and Link, T. E. (2013). A methodology to characterize vertical accuracies in lidar-derived products at landscape scales. *Photogrammetric Engineering & Remote Sensing*, 79(8):709–716
- Rott, H., Nagler, T., Prinz, R., Voglmeier, K., Fromm, R., Adams, M. S., Yueh, S., Elder, K., **Marshall, H.P.**, Coccia, A., Imbembo, E., Schuttemeyer, D., and Kern, M. (2013). AlpSAR 2012-13, a field experiment on snow observations and parameter retrievals with Ku-and X-band radar. *ESA Living Planet Symposium*, ESA SP-722.2-13:92
- 2012 Mikesell, T. D., van Wijk, K., Haney, M. M., Bradford, J. H., **Marshall, H.P.**, and Harper, J. T. (2012). Monitoring glacier surface seismicity in time and space using rayleigh waves. *Journal of Geophysical Research-Earth Surface*, 117(F2):F02020
- 2011 **Marshall, H.P.**, Schweizer, J., and Birkeland, K. (2011). Some recent advances in snow and avalanche science. *Cold Regions Science and Technology*, 69(2-3):119–121
- Geroy, I., Gribb, M., **Marshall, H.P.**, Chandler, D., Benner, S. G., and McNamara, J. P. (2011). Aspect influences on soil water retention and storage. *Hydrological Processes*, 25(25):3836–3842
- 2010 Tape, K. D., Rutter, N., **Marshall, H.P.**, Essery, R., and Sturm, M. (2010b). Recording microscale variations in snowpack layering using near-infrared photography. *Journal of Glaciology*, 56(195):75–80
- Tape, K. D., Lord, R., **Marshall, H.P.**, and Ruess, R. W. (2010a). Snow-mediated ptarmigan browsing and shrub expansion in arctic alaska. *Ecoscience*, 17(2):186–193

- Koh, G., Lever, J., Arcone, S., **Marshall, H.P.**, and Ray, L. (2010). Autonomous fmcw radar survey of antarctic shear zone. *IEEE Publications, 13th International Conference on Ground Penetrating Radar*, DOI: 10.1109/ICGPR.2010.5550174
- 2009 **Marshall, H.P.** and Johnson, J. B. (2009). Accurate inversion of high-resolution snow penetrometer signals for microstructural and micromechanical properties. *Journal of Geophysical Research-Earth Surface*, 114(F04016)
- Lutz, E., Birkeland, K., and **Marshall, H.P.** (2009). Quantifying changes in weak layer microstructure associated with artificial load changes. *Cold Regions Science and Technology*, 59(2-3):202–209
- Pielmeier, C. and **Marshall, H.P.** (2009). Rutschblock-scale snowpack stability derived from multiple quality-controlled snowmicropen measurements. *Cold Regions Science and Technology*, 59(2-3):178–184
- 2008 **Marshall, H.P.** and Koh, G. (2008). FMCW radars for snow research. *Cold Regions Science and Technology*, 52(2):118–131
- Hardy, J., Davis, R., Koh, Y., Cline, D., Elder, K., Armstrong, R., **Marshall, H.P.**, Painter, T., Saint-Martin, G. C., Deroo, R., Sarabandi, K., Graf, T., Koike, T., and McDonald, K. (2008). NASA Cold Land Processes Experiment (CLPX 2002/03): Local Scale Observation Site. *Journal of Hydrometeorology*, 9(6):1434–1442
- Marshall, H.P.**, Koh, G., and Sturm, M. (2008). Ultra-broadband portable microwave fmcw radars for measuring snow depth, snow water equivalent, and stratigraphy: practical considerations. *Proceedings of the 29th International Union of Radio Science (URSI) General Assembly*, page DOI:10.1.1.498.5307. (Invited)
- 2007 **Marshall, H.P.**, Schneebeli, M., and Koh, G. (2007). Snow stratigraphy measurements with high-frequency FMCW radar: Comparison with snow micro-penetrometer. *Cold Regions Science and Technology*, 47(1-2):108–117
- O’Neel, S., **Marshall, H.P.**, McNamara, D. E., and Pfeffer, W. T. (2007). Seismic detection and analysis of icequakes at Columbia Glacier, Alaska. *Journal of Geophysical Research-Earth Surface*, 112(F3)
- Waddington, E. D., Neumann, T. A., Koutnik, M. R., **Marshall, H.P.**, and Morse, D. L. (2007). Inference of accumulation-rate patterns from deep layers in glaciers and ice sheets. *Journal of Glaciology*, 53(183):694–712
- 2005 **Marshall, H.P.**, Koh, G., and Forster, R. R. (2005). Estimating alpine snowpack properties using FMCW radar. *Annals of Glaciology*, 40:157–162
- 2004 **Marshall, H.P.**, Koh, G., and Forster, R. R. (2004b). Ground-based frequency-modulated continuous wave radar measurements in wet and dry snowpacks, Colorado, USA: an analysis and summary of the 2002-03 NASA CLPX data. *Hydrological Processes*, 18(18):3609–3622
- 2002 **Marshall, H.P.**, Harper, J. T., Pfeffer, W. T., and Humphrey, N. F. (2002). Depth-varying constitutive properties observed in an isothermal glacier. *Geophysical Research Letters*, 29(23):61–1:61–4

- 1999 **Marshall, H.P.**, Conway, H., and Rasmussen, L. A. (1999). Snow densification during rain. *Cold Regions Science and Technology*, 30(1-3):35–41
- Conway, H., Rasmussen, L. A., and **Marshall, H.P.** (1999). Annual mass balance of Blue Glacier, USA: 1955-97. *Geografiska Annaler, Ser. A-Phys. Geogr.*, 81A(4):509–520
- Morse, D. L., Waddington, E. D., **Marshall, H.P.**, Neumann, T. A., Steig, E. J., Dibb, J. E., Winebrenner, D. P., and Arthern, R. J. (1999). Accumulation rate measurements at Taylor Dome, East Antarctica: Techniques and strategies for mass balance measurements in polar environments. *Geografiska Annaler, Ser. A-Phys. Geogr.*, 81A(4):683–694

Refereed Book Chapters / Technical Reports

- 2021 Arenson, L., Colgan, W., and **Marshall, H.P.** (2021, 2nd edition). *Physical, Thermal, and Mechanical Properties of Snow, Ice, and Permafrost*, chapter 2, pages 35–75. Snow and ice-related hazards, risks, and disasters. Elsevier, ed. Haeberli and Whiteman, Radarweg 29, PO Box 211, 1000 AE Amsterdam, Netherlands
- Youngblood, P. and **Marshall, H.P.** (2021). Continuous snow temperature monitoring for the idaho highway 21 avalanche program. Technical Report FHWA-ID-21-276, Idaho Transportation Department, 3311 W State Street, P. O. Box 7129, Boise, ID 83707-1129 United States
- Johnson, J., **Marshall, H.P.**, and Anderson, J. (2021b). Infrasound Avalanche Monitoring: Enhanced monitoring in Little Cottonwood Canyon, Utah. Digital media, CDOT-2021-10, Colorado Department of Transportation
- 2019 Meehan, T. G., **Marshall, H.P.**, Deeb, E. J., and Shoop, S. A. (2019). Snowmicropenetrometer applications for winter vehicle mobility. ERDC 6.2 Boreal Aspects of Ensured Maneuver (BAEM) TR-19-14, U.S. Army Cold Regions Research and Engineering Laboratory
- 2015 **Marshall, H.P.**, Hawley, R., and Tedesco, M. (2015). *Field measurements for remote sensing of the cryosphere*, chapter 14, pages 345–381. Remote Sensing of the Cryosphere. John Wiley & Sons, Ltd, ed. M. Tedesco
- Arenson, L., Colgan, W., and **Marshall, H.P.** (2015). *Physical, Thermal, and Mechanical Properties of Snow, Ice, and Permafrost*, chapter 2, pages 35–75. Snow and ice-related hazards, risks, and disasters. Elsevier, ed. Haeberli and Whiteman, Radarweg 29, PO Box 211, 1000 AE Amsterdam, Netherlands
- 2014 Havens, S., **Marshall, H.P.**, Trisca, G., and Johnson, J. B. (2014b). Real time avalanche detection for high risk areas. Monograph FHWA-ID-14-219, Idaho Transportation Department, 3311 W State Street, P. O. Box 7129, Boise, ID 83707-1129 United States

Peer-Reviewed Archived Datasets [peer-reviewed at NSIDC and Arctic Data Center=33]

- 2024 Ciafone, S., O'Neel, S., Adebisi, N., Zikan, K., Enterkine, J., van der Weide, T., Wilder, B., Hoppinen, Z., Filiano, D., and **Marshall, H.P.** (2024a). SnowEx Mores Creek Summit (MCS) Airborne LiDAR Survey Raw, Version 1. Digital media, <https://doi.org/10.5067/OYF98UGSOUQY>, NASA National Snow and Ice Data Center Distributed Active Archive Center
- Ciafone, S., O'Neel, S., Adebisi, N., Zikan, K., Enterkine, J., van der Weide, T., Wilder, B., Hoppinen, Z., Filiano, D., and **Marshall, H.P.** (2024b). SnowEx Mores Creek Summit (MCS) Airborne LiDAR Survey, Version 1. Digital media, <https://doi.org/10.5067/NE8CLN3ES5XQ>, NASA National Snow and Ice Data Center Distributed Active Archive Center

Hoppinen, Z. and **Marshall, H.P.** (2024). SnowEx UAVSAR L-band Synthetic Aperture Radar, Version 1. Digital media, https://nsidc.org/data/snex_uavsar/versions/1, NASA National Snow and Ice Data Center Distributed Active Archive Center

Stuefer, S., May, L., Mason, M., Vuyovich, C., **Marshall, H.P.**, Vas, D., and Elder, K. (2024b). Snowex23 snow water equivalent, version 1. Digital media, <https://doi.org/10.5067/FRAMQ2JFF6J9>, NASA National Snow and Ice Data Center Distributed Active Archive Center

May, L., Stuefer, S., Vuyovich, C., Elder, K., Osmanoglu, B., **Marshall, H.P.**, Durand, M., Vas, D., Gelvin, A., Liddle Broberg, K., and Maakestad, J. (2024). SnowEx23 Mar22 IOP Snow Depth Measurements, Version 1. Digital media, <https://doi.org/10.5067/086OMZDJP2W6>, NASA National Snow and Ice Data Center Distributed Active Archive Center

Mason, M., **Marshall, H.P.**, McCormick, M., Craaybeek, D., Elder, K., and Vuyovich, C. (2024). SnowEx20 Time Series Snow Pit Measurements, Version 2. Digital media, <https://doi.org/10.5067/KZ43HVLZV6G4>, NASA National Snow and Ice Data Center Distributed Active Archive Center

Stuefer, S., May, L., Bailey, J., Vas, D., Mason, M., Hale, K., **Marshall, H.P.**, Vuyovich, C., Elder, K., and SnowEx Alaska March 2023 Team (2024a). Snowex23 mar23 iop community snow depth measurements, version 1. Digital media, <https://doi.org/10.5067/6QD3UJVABY6D>, NASA National Snow and Ice Data Center Distributed Active Archive Center

Brangers, I., De Lannoy, G., **Marshall, H.P.**, Dunmire, D., Bonnell, R., Cox, B., Capelle, J., Baxter, B., and Lievens, H. (2024a). SnowEx23 Mar23 Mobile Tower-Based Dual-Pol C-Band, Version 1. Digital media, <https://doi.org/10.5067/9BTGU6GJXTQT>, NASA National Snow and Ice Data Center Distributed Active Archive Center

Hale, K., **Marshall, H.P.**, Stuefer, S., and Vuyovich, C. (2024). SnowEx23 Laser Snow Microstructure Specific Surface Area Snow-off Data, Version 1. Digital media, <https://doi.org/10.5067/9SY1H2L0BY0X>, NASA National Snow and Ice Data Center Distributed Active Archive Center

Meloche, J., Lemmetyinen, J., Meyer, K., Alabi, I., Vuyovich, C., Stuefer, S., **Marshall, H.P.**, Durand, M., and Langlois, A. (2023a). SnowEx23 Laser Snow Microstructure Specific Surface Area Data, Version 1. Digital media, <https://doi.org/10.5067/BSEP59ADC6XN>, NASA National Snow and Ice Data Center Distributed Active Archive Center

2023 Mason, M., **Marshall, H.P.**, McCormick, M., Craaybeek, D., Elder, K., and Vuyovich, C. M. (2023). Snowex20 time series snow pit measurements, version 1. Digital media, doi:10.5067/POT9E0FFUUD1, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA

Meloche, J., Lemmetyinen, J., Meyer, K., Alabi, I., Vuyovich, C. M., Stuefer, S., **Marshall, H.P.**, Durand, M., and Langlois, A. (2023b). SnowEx23 laser snow microstructure specific surface area data, version 1. Digital media, doi: 10.5067/BSEP59ADC6XN, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA

- Adebisi, N., **Marshall, H.P.**, O'Neel, S., Vuyovich, C. M., Hiemstra, C., and Elder, K. (2023a). Snowex20-21 qsi lidar dem 3m utm grid, version 1. Digital media, doi: 10.5067/VV382L4MQI7V, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- Adebisi, N., **Marshall, H.P.**, Vuyovich, C. M., Hiemstra, C., Elder, K., and Durand, M. (2023b). SnowEx20-21 QSI lidar snow depth 0.5m utm grid, version 1. Digital media, doi: 10.5067/VBUN16K365DG, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- Adebisi, N., **Marshall, H.P.**, Vuyovich, C. M., Hiemstra, C., Elder, K., and Durand, M. (2023c). SnowEx20-21 QSI lidar snow depth 3m utm grid, version 1. Digital media, doi:10.5067/4VO7438DP332, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- Adebisi, N., **Marshall, H.P.**, Vuyovich, C. M., Hiemstra, C., Elder, K., and Durand, M. (2023d). SnowEx20-21 QSI lidar vegetation height 0.5m utm grid, version 1. Digital media, doi: 10.5067/8RBIUIPEUJ7Z, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- Adebisi, N., **Marshall, H.P.**, Vuyovich, C. M., Hiemstra, C., Elder, K., and Durand, M. (2023e). SnowEx20-21 QSI lidar vegetation height 3m utm grid, version 1. Digital media, doi: 10.5067/K67GKZQ9WUEI, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- 2022 Breen, C. M., Lumbrazo, C., Hiemstra, C., Vuyovich, C. M., Raleigh, M. S., and **Marshall, H.P.** (2022). Snowex20 time-lapse imagery, version 1. Digital media, <https://doi.org/10.5067/P3D1QRH7O8X5>, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- Adebisi, N., **Marshall, H.P.**, O'Neel, S., Vuyovich, C. M., Hiemstra, C., and Elder, K. (2022). Snowex20-21 qsi lidar dem 0.5m utm grid, version 1. Digital media, doi: 10.5067/YO583L7ZOLOO, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- McDowell, I., Keegan, K., Osterberg, E., Hawley, R., and **Marshall, H.P.** (2022). Grain size and ice layer stratigraphy from the upper 10 meters of cores 1-7 collected during the 2016 greenland traverse for accumulation and climate studies (greentracs), southwestern greenland. Digital media,, Arctic Data Center
- Houser, P., Rudisill, W., Johnston, J., Elder, K., **Marshall, H.P.**, Vuyovich, C. M., Kim, E. J., and Mason, M. (2022). Snowex meteorological station measurements from grand mesa, co, version 1. Digital media, DOI:10.5067/497NQVJ0CBEX, NASA National Snow and Ice Data Center Distributed Active Archive Center
- 2021 Brucker, L., Vuyovich, C. M., Elder, K., **Marshall, H.P.**, and Hiemstra, C. (2021). SnowEx20 Grand Mesa autumn 2019 snow pits, version 1. Digital media, doi: <https://doi.org/10.5067/4IZGCQC1J31Q>, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA

- Hiemstra, C., Vuyovich, C. M., and **Marshall, H.P.** (2021). SnowEx20 Grand Mesa Reference GIS data sets, version 1. Digital media, doi: <https://doi.org/10.5067/YDZXY4Q79VIJ>, NASA National Snow and Ice Data Center Distributed Active Archive Center
- Vuyovich, C. M., **Marshall, H.P.**, Elder, K., Hiemstra, C., Brucker, L., and McCormick, M. (2021). Snowex20 grand mesa intensive observation period snow pit measurements, version 1. Digital media, doi: <https://doi.org/10.5067/DUD2VZEVBJS>, NASA National Snow and Ice Data Center Distributed Active Archive Center
- McGrath, D., Bonnell, R., Webb, R., and **Marshall, H.P.** (2021). SnowEx20 Grand Mesa IOP CSU 1GHz GPR raw, version 1. Digital media, doi: [10.5067/CT6NS2LIASRS](https://doi.org/10.5067/CT6NS2LIASRS), NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- Bonnell, R., McGrath, D., Webb, R., and **Marshall, H.P.** (2021a). SnowEx20 Grand Mesa IOP CSU 1GHz GPR, version 1. Digital media, doi: [10.5067/S5EGFLCIAB18](https://doi.org/10.5067/S5EGFLCIAB18), NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- 2020 **Marshall, H.P.** and Gleason, A. (2020). SnowEx17 Senator Beck SnowMicroPen (SMP) Raw Penetration Force Profiles, Version 1. Digital media, doi:<https://doi.org/10.5067/K7P8PB6EO1HO>, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- Hiemstra, C., **Marshall, H.P.**, Vuyovich, C. M., Elder, K., Mason, M., and Durand, M. (2020). SnowEx20 Community Snow Depth Probe Measurements, Version 1. Digital media, doi: <https://doi.org/10.5067/9IA978JIACAR>, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- Mason, M., **Marshall, H.P.**, and Merkouriadi, I. (2020). SnowEx20 SnowMicroPen (SMP) Raw Penetration Force Profiles, Version 1. Digital media, doi: <https://doi.org/10.5067/ZYW6IHFRYDSE>, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- Johnson, J., Anderson, J., **Marshall, H.P.**, Havens, S., and Watson, L. (2020). Dataset on fortress snow avalanche 13 march 2018. *Boise State University Infrasound Repository*, 4
- 2018 Elder, K., Brucker, L., Hiemstra, C., and **Marshall, H.P.** (2018). Snowex17 community snow pit measurements, version 1. Digital media, <https://doi.org/10.5067/Q0310G1XULZS>, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- Brucker, L., Hiemstra, C., **Marshall, H.P.**, and Elder, K. (2018a). Snowex17 community snow depth probe measurements, version 1. Digital media, <https://doi.org/10.5067/WKC6VFMT7JTF>, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- 2004 **Marshall, H.P.**, Koh, G., and Forster, R. (2004a). CLPX-Ground: Ground-based Frequency Modulated Continuous Wave (FMCW) radar. Digital media, <http://nsidc.org/data/nsidc-0164.html>, National Snow and Ice Data Center, University of Colorado at Boulder

Magazine Articles / Podcasts

- | | |
|------|--|
| 2013 | Durham, T., Marshall, H.P. , Tsang, L., Racette, P., Bonds, Q., Miranda, F., and Vanhille, K. (2013). Wideband sensor technologies for measuring surface snow. <i>Earthzine</i> , December 2(earthzine.org) |
| 2012 | Marshall, H.P. (2012). Researching snow: More than just hitting the slopes. Podcast https://beyondtheblue.boisestate.edu/blog/2012/01/09/hp-marshall/ , Beyond the Blue: Boise State Faculty Podcasts |
| 2011 | Sturm, M. and Marshall, H.P. (2011). Marshall receives Cryosphere Young Investigator Award (citation and response). <i>EOS Transactions, AGU</i> , 92(22):189 |

Research Funding Summary [chronological order]

Agency	Program, lead institution (if not BSU)	Role	Dates	Total Award	BSU Award
NSF	Arctic Sciences	PI	9/2005-8/2010	\$69,785	\$0
NASA	Terrestrial Hydrology, NEWS	PI	8/2006-7/2009	\$462,132	\$114,232
NASA	Terrestrial Hydrology Program (THP)	PI	8/2010-8/2012	\$35,468	\$35,468
NASA	Instrument Incubator Program (IIP)	PI	11/2011-12/2014	\$4,000,000	\$94,947
ITD	Transportation Research Program	PI	9/2010-8/2011	\$10,000	\$10,000
Oil industry	DF Dickins Associates and consortium	PI	1/2010-12/2012	\$328,747	\$328,747
NASA	Earth Systems Science Fellowship (ESSF), Scott Havens	PI	9/2010-12/2013	\$90,000	\$90,000
NSF	Hydrologic Sciences	PI	4/2010-3/2013	\$228,963	\$228,963
NASA	New Investigator Program (NIP)	PI	9/2010-8/2013	\$329,953	\$329,953
NASA	Experimental Program to Stimulate Competitive Research (EPSCoR)	PI	1/2011-12/2013	\$1,124,552	\$749,620
ITD	Idaho Transportation Department (ITD)	PI	9/2012-8/2015	\$100,000	\$100,000
NSF	RAPID	Co-PI	4/2012-4/2013	\$19,912	\$19,912
NSF	Major Research Instrumentation (MRI)	Co-PI	9/2012-8/2013	\$336,122	\$336,122
NSF	Major Research Instrumentation (MRI)	Co-PI	9/2012-8/2013	\$555,384	\$555,384
OGP	International Association of Oil and Gas Producers (OGP)	Co-PI	1/2014-12/2014	\$1,409,449	\$128,731
NASA	Terrestrial Hydrology Program (THP)	Co-PI	1/2014-12/2014	\$65,340	\$65,340
DoD	US Army Basic Research Program	Co-PI	10/2013-9/2016	\$490,800	\$0
NASA	Experimental Program to Stimulate Competitive Research (EPSCoR)	Co-PI	6/2014-6/2018	\$749,870	\$749,870
NASA	Terrestrial Hydrology Program (THP)	Co-PI	7/2015-6/2018	\$295,577	\$295,577
NSF	Office of Polar Programs (OPP)	PI	1/2015-12/2017	\$1,000,000	\$355,338
NASA	Instrument Incubator Program (IIP)	PI	2/2015-5/2017	\$3,000,000	\$330,000
TARP	U.S. Department of Transportation Avalanche Research Program (TARP)	PI	6/2017-8/2018	\$99,339	\$99,339
JPL	NASA Jet Propulsion Laboratory	PI	6/2017-8/2018	\$21,103	\$21,103
NASA	Terrestrial Hydrology Program (THP-16), CRREL	Co-PI	6/2017-6/2022	\$450,000	\$75,736
NASA	Terrestrial Hydrology Program (THP-16)	PI	6/2017-6/2022	\$460,719	\$460,719
NASA	Terrestrial Hydrology Program (THP-17), CRREL	Co-PI	6/2018-6/2021	\$421,212	\$32,228
NASA	Terrestrial Hydrology Program (THP-17), CSU	Co-PI	6/2018-6/2021	\$444,742	\$230,396
TARP	U.S. Department of Transportation Avalanche Research Program (TARP)	Co-PI	5/2019-12/2021	\$99,509	\$99,509
ITD	Idaho Transportation Department (ITD)	PI	9/2018-12/2020	\$45,000	\$45,000
ID Dept Commerce	Idaho Global Entrepreneurial Mission (IGEM)	Co-PI	6/2018-7/2020	\$368,772	\$368,772
DoD	US Army Cold Regions Research and Engineering Lab (CRREL) BAA	PI	9/2020-8/2022	\$1,000,000	\$1,000,000
NASA	Terrestrial Hydrology Program (THP)	PI	12/2020-8/2022	\$419,795	\$419,795
NASA	NASA Earth and Space Science Fellowship (NESSF), Joel Gongora	PI	9/2018-8/2023	\$133,864	\$133,864
JPL	NASA Jet Propulsion Laboratory	PI	1/2023-9/2023	\$60,366	\$60,366
NASA	Experimental Program to Stimulate Competitive Research (EPSCoR)	Co-PI	1/2021-12/2023	\$791,669	\$527,301
NSF	Major Research Instrumentation (MRI)	PI	9/2018-8/2024	\$1,615,385	\$161,883
NASA	Instrument Incubator Program (IIP-19)	PI	2/2020-5/2024	\$2,168,639	\$1,123,384
USFS	Rocky Mountain Research Station	PI	10/2021-9/2024	\$62,442	\$62,442
NSF	Arctic Observing Network (AON), WSU	Co-PI	1/2022-11/2024	\$1,833,668	\$204,068
NASA	Terrestrial Hydrology Program (THP-20), UWYoming	Co-PI	1/1/2022-12/2024	\$465,176	\$48,133
NASA	Terrestrial Hydrology Program (THP), SnowEx	PI	6/2022-9/2024	\$137,968	\$137,968
NASA	Instrument Incubator Program (IIP-21, UMass Amherst)	Co-PI	1/2022-2/2025	\$4,497,426	\$354,698
DoD CRREL	UCSB/DRI/LBNL/BSU TriState Snow Consortium	PI	9/2023-8/2025	\$4,500,000	\$1,500,000
NASA	Future Inv NASA Earth & Space Science and Tech (FINESST), Zach Hoppinen	PI	9/2023-8/2026	\$148,643	\$148,643
DoD CRREL	Broad Agency Announcement	PI	7/2024-7/2027	\$3,284,205	\$1,939,225
Belgian NSF	Earth Observation	Co-PI	9/2024-9/2029		\$108,000
US DOI	Bureau of Reclamation	PI	2/2024-2/2027		\$460,000
NASA	Terrestrial Hydrology Program (THP-24), UCLA	Co-PI	6/2025-5/2028		\$105,900
NASA	Terrestrial Hydrology Program (THP-24), UAF	Co-PI	6/2025-5/2028		\$86,116
NASA	Terrestrial Hydrology Program (THP-24), Goddard	Collaborator	6/2025-5/2028		
NASA	Terrestrial Hydrology Program (THP-24), CRREL	Collaborator	6/2025-5/2028		
active awards				\$16,565,182	\$5,092,751
total				\$39,929,792	\$14,932,792

Awarded Research Grants [reverse chronological order]

- 6/2025-5/2028 *Development of a multi-frequency InSAR phase data assimilation framework for improved seasonal SWE characterization*
Co-PI, \$105,900 to BSU
PI Team: Steve Margulis (UCLA), Rashmi Shaw and Xiaolan Xu (JPL)
- 6/2025-5/2028 *SnowEx Alaska synthesis: deriving and validating spatially distributed SWE from NASA's SWESARR using in-situ, airborne, and modeled data*
Co-PI, \$86,116 to BSU
PI Team: Kate Hale and Sveta Stuefer (University Alaska Fairbanks), Jack Tarricone and Carrie Vuyovich (NASA Goddard)
- 6/2025-5/2028 *NISAR for global snow water equivalent: Evaluating L-band InSAR for basin scale snow monitoring by leveraging airborne lidar, modeling, and the SnowEx campaigns*
Collaborator, \$0 to BSU
PI Team: Jack Tarricone and Carrie Vuyovich (NASA Goddard), Kelly Elder USFS, Randall Bonnell (USGS), Mark Robertson (M3Works), Tom Painter (ASO)
- 6/2025-5/2028 *A C- and L-band Fusion for Global NISAR SWE Retrievals*
Collaborator, \$0 to BSU
PI Team: Zach Hoppinen (CRREL) and Ross Palomaki (University Colorado Boulder)
- 2/2024-2/2027 *Development of a Snow Water Equivalent "Landing Strip" for Use Towards Improved CBRFC Water Supply Forecasts*
PI, \$460,000 to BSU
PI Team: Joe Meyer (BSU)
- 9/2024-9/2029 *Water and vegetation tower radar experiments for improved climate monitoring – WAVETRAX*
Co-PI, \$108,000 to BSU
PI Team: Hans Lievens (Ghent University)
- 7/2024-7/2027 DoD U.S. Army Cold Regions Research and Engineering Lab (CRREL)
Snow strength mapping for mobility applications: combining field observations, radar remote sensing, and modeling to create local to global snow trafficability products
PI, \$3,284,205 total, \$1,939,225 to BSU
PI Team: Glen Liston (Colorado State University) and Kelly Elder (USFS Rocky Mountain Research Station)
- 9/2023-8/2026 Future Investigators in NASA Earth and Space Science and Technology (FINESST)
Evaluating NISAR's InSAR Potential in Snow-Covered Regions
PI, \$148,643
Student: Zach Hoppinen
- 9/2023-8/2025 DoD U.S. Army Cold Regions Research and Engineering Lab (CRREL)
Climate and natural hazards, snow-covered and mountain environment sensing research
Institutional PI, \$4,500,000 total, \$1,500,000 to BSU
PI Team: Jeff Dozier and Ned Bair (Univ California Santa Barbara, overall lead), Sean McKenna (Desert Research Institute), Erica Siirila-Woodburn (Lawrence Berkeley National Laboratory)
- 1/2022-2/2025 NASA Instrument Incubator Program (IIP-21)

Snow Water-equivalent Wide Swath SAR and InSAR (SNOWWI)

Institutional PI, \$4,497,426 total, \$354,698 to BSU

PI Team: Paul Siqueira (Univ Mass, overall lead), Eli Deeb (U.S. Army CRREL)

6/2022-9/2024 NASA Terrestrial Hydrology Program (THP)

SnowEx 2022-23 Leadership Team

PI, \$137,968 to BSU

Leadership Team: Carrie Vuyovich (NASA Goddard), Sveta Stuefer (UAF), Kelly Elder (USFS), Mike Durand (OSU)

1/2022-12/2024 NASA Terrestrial Hydrology Program (THP)

Evaluation of L-band InSAR using L-band GPR: Sensitivity to Liquid Water Content, Topography, and Land Cover to Determine Feasibility for Global SWE Applications

Institutional PI, \$465,176 total, \$48,133 to BSU

PI Team: Ryan Webb (Univ WY, overall lead), Dan McGrath (CSU)

1/2022-11/2024 NSF Arctic Observing Network (AON)

The Integrated Characterization of Energy, Clouds, Atmospheric state, and Precipitation at Summit (ICECAPS) - Measurements along Lagrangian Transects (MELT)

Institutional PI, \$3,531,764 total, \$204,068 to BSU

PI Team: Von Walden (WSU, overall lead), Matthew Shupe (CU Boulder), Bob Hawley (Dartmouth), Ralf Bennartz (Univ Wisconsin-Madison)

10/2021-9/2024 *Analysis of Snow Thermal Data*

PI, \$62,442

Collaborator: Charlie Luce (USFS)

2/2020-1/2023 NASA Instrument Incubator Program

Digitally Enhanced Meta-surface Radar/Radiometer for Snow Remote Sensing

Lead PI, \$2,168,639 total, \$1,123,384 to BSU

PI Team: Adrian Tang (JPL), Frank Chang (UCLA)

9/2018-8/2022 NSF: Major Research Instrumentation

Development of a Long-Range UAV For Sea Ice Monitoring

Co-PI, \$1,615,385 total, \$161,883 to BSU

PI team: Andy Mahoney (UAF, overall lead), CoPIs: Hajo Eiken, Dejan Raskovic, Denise Thorsen, Michael Hatfield (UAF)

1/2021-12/2023 NASA Experimental Program to Stimulate Competitive Research (EPSCoR)

CryoIdaho: Building Idaho's Cryosphere Research Community through Analysis of Terrain Effects on Seasonal Accumulation and Melt of Snow and Ice

Co-PI, \$791,669 total, \$527,301 to BSU

PI: Ellyn Enderlin, CoPIs: Michal Kopera, Lejo Flores, Jim McNamara (BSU), Tim Bartholomaeus (Univ Idaho)

1/2023-9/2023 NASA Jet Propulsion Laboratory / CalTech

Signals of Opportunity: Grand Mesa 2023 Campaign

PI, \$60,366

09/2018-8/2023 NASA Earth and Space Science Fellowship

Optimal Sampling Strategies for Snow: Application of Machine Learning and Graph Theory

PI, \$133,864

Student: Joel Gongora

12/2020-8/2022 NASA Terrestrial Hydrology Program (THP)

SnowEx 2020-2021 LiDAR

PI, \$419,795

9/2020-8/2022 U.S. Army Cold Regions Research and Engineering Laboratory BAA

Advancement of snow monitoring for water resources, vehicle mobility, and hazard mitigation: using optical, microwave, acoustic, and seismic techniques

PI, \$1,000,000

Co-PIs: Ellyn Enderlin, Dylan Mikesell, Lee Liberty, Jeff Johnson, Jake Anderson

7/2019-12/2021 Consortium of U.S. Dept. of Transportation Avalanche Forecasting Programs

Infrasound Avalanche Monitoring: Enhanced Monitoring in Little Cottonwood Canyon, Utah

Co-PI, \$99,509

PI: Jeff Johnson (BSU)

6/2018-7/2020 Idaho Department of Commerce

A General Purpose Goniometer

Co-PI, \$368,772

PI: Sing Ming Loo (BSU)

09/2018-12/2020 Idaho Department of Transportation Research Program

Continuous snow temperature monitoring at Banner Summit, Idaho

PI, \$45,000

6/2018-6/2021 NASA Terrestrial Hydrology Program 2017

Constraining airborne and satellite algorithms for estimating snow water equivalent through spatially extensive ground penetrating radar observations during NASA's SnowEx campaign

Co-PI, \$444,742 total, \$230,396 to BSU

PI: Dan McGrath (Colorado State Univ.), Co-PI: Ryan Webb (Univ. Colorado)

6/2018-6/2021 NASA Terrestrial Hydrology Program 2017

Using SnowEx Year 1 Data to Assess and Improve Snow Measurements, Validation, Algorithms, and Data Assimilation in Forest and Shrubland Landscapes

Co-PI, \$421,212 total, \$32,228 to BSU

PI: Chris Hiemstra (CRREL), Co-PIs: Kelly Elder (USFS), Ludo Brucker (NASA Goddard)

6/2017-6/2022 NASA Terrestrial Hydrology Program 2016

Spatiotemporal patterns in ground-based snow remote sensing and in-situ observations: Analysis of CLPX-II, CoreH2O, and SnowEx Y1 data for sampling design and site selection

PI, \$460,719

Co-PIs: Kelly Elder (USFS), Eli Deeb (CRREL), Ludo Brucker (NASA Goddard)

6/2017-6/2022 NASA Terrestrial Hydrology Program 2016

Remote sensing strategies and requirements for L-Band interferometric snow depth and snow water equivalent estimation

Co-PI, \$450,000 total, \$75,736 to BSU

PI: Eli Deeb (CRREL), Co-PIs: Rick Forster (Univ. Utah), Cathleen Jones (NASA JPL), Chris Hiemstra (CRREL)

6/2017-8/2018 NASA Jet Propulsion Laboratory, CalTech

ASO and UAVSAR Data Fusion: Exploring the Advantages of Combining Airborne LIDAR and InSAR

PI, \$21,103

6/2017-8/2018 Consortium of U.S. Department of Transportation Avalanche Forecasting Programs

Infrasound Avalanche Detection: Comparison Study of Multiple New Technologies

PI, \$99,339

Co-PIs: Jeff Johnson, Sin Ming Loo (BSU)

2/2015-5/2017 NASA Instrument Incubator Program (IIP) - Phase 2

Broadband Array Technology in Support of the Snow and Cold Land Process (SCLP) Mission

PI, \$3,000,000 total, \$330,000 to BSU

1/2015-12/2017 NSF Office of Polar Programs

Collaborative Research: GreenTrACS: Greenland Traverse for Accumulation and Climate Studies

PI, \$1,000,000 total, \$355,338 to BSU

Co-PI: John Bradford (BSU)

7/2015-6/2018 NASA Terrestrial Hydrology Program (THP)

Multiple Frequency Active Microwave Remote Sensing for Snow Water Equivalent Retrieval from Space: A Data Assimilation Approach

Co-PI, \$295,577

PI: Lejo Flores (BSU)

6/2014-6/2018 NASA Experimental Program to Stimulate Competitive Research (EPSCoR)

Monitoring Earth's Hydrosphere: Integrating Remote Sensing, Modeling, and Verification

Co-PI, \$749,870

PI: Lejo Flores, Co-PIs: Jim McNamara, Nancy Glen (BSU)

10/2013-9/2016 U.S. Army Basic Research Program

Quantifying the accuracy of Snow Water Equivalent estimates using radar phase over a wide range of frequencies and varying snow conditions

Co-PI, \$490,800, expert consultant with CRREL

PI: Eli Deeb (CRREL), Co-PI: Steve Arcone (CRREL)

1/2014-12/2014 NASA Terrestrial Hydrology Program (THP)

NASA Snow School

Co-PI, \$65,340

PI: Mark Sereze (National Snow and Ice Data Center), Co-PIs: Michael Durand (Ohio State University), Kelly Elder (USFS), Jessica Lundquist (University of Washington), Noah Molotch (University of Colorado), Matthew Sturm (University of Alaska, Fairbanks)

01/2014-12/2014 International Association of Oil and Gas Producers (OGP)

Program Design and Management for Remote Sensing Tests with Oil in Ice

Co-PI, \$1,409,449 total, BSU portion: \$128,731

BSU PI: John Bradford

Collaborators: Oil Spill Recovery Institute, University of Alaska Fairbanks, Prince William Sound Science Center, U.S. Army Cold Regions Research and Engineering Laboratory (CRREL), Polar Ocean Services, Woods Hole Oceanographic Institute, Norbit, Inland Gulf Marine, University of Washington

09/2012-08/2013 NSF Major Research Instrumentation

MRI: Acquisition of a GPU-accelerated High Performance Computing and Visualization Cluster

Co-PI, \$555,384

PI: Inanc Senocak, Co-PIs: Peter Mullner, Jodi Mead, Tim Anderson (BSU)

09/2012-08/2013 NSF Major Research Instrumentation

MRI: Development of a laser-ultrasonic ice core tomography system

Co-PI, \$336,122

PI: Kasper Van Wijk (BSU)

04/2012-04/2013 NSF Hydrologic Sciences

RAPID: An unusual opportunity to track snow ablation using stable isotope evolution of the 2011-2012 snowpack near Boise, Idaho

Co-PI, \$19,912

PI: Sam Evans, Co-PIs: Alejandro Flores, Matt Kohn (BSU)

09/2012-08/2015 Idaho Department of Transportation

Development of an operational infrasonic avalanche detection system for Highway 21, Idaho

PI, \$100,000

Co-PIs: Scott Havens, Jeff Johnson (BSU)

01/2011-12/2013 NASA Experimental Program to Stimulate Competitive Research (EPSCoR)

Remote sensing of the Cryosphere: calibration and validation

Science PI, \$1,124,552 (\$374,932 university match)

Co-PI: John Bradford, Kasper Van Wijk, Jim McNamara, and Alejandro Flores, BSU Geosciences (BSU)

09/2010-08/2013 NASA New Investigator Program

Improvement of remotely sensed snow water equivalent estimates from microwave radar

PI, \$329,953 (1st in Idaho history)

04/2010-03/2013 NSF Hydrological Sciences Program

Quantifying lateral flow of water in alpine snowpacks using high-resolution geophysical techniques

PI, \$228,963

Co-PI: John Bradford and Jim McNamara, BSU

09/2010-12/2013 NASA Earth and Space Science Fellowship

Characterizing snow microstructure with X-ray tomography and a snow micropenetrrometer: understanding important effects of microstructure on remote sensing

PI, \$90,000

Student: Scott Havens

01/2010-12/2012 DF Dickins Associates and consortium of oil companies

Detecting oil on and under sea ice using ground-penetrating radar: proposal for developing a new airborne system

PI, \$328,747

Co-PI: John Bradford, BSU

09/2010-08/2011 Idaho Department of Transportation Research Program

Development of an avalanche forecasting model for Highway 12 and 21

PI, \$10,000

11/2011-12/2014 NASA Instrument Incubator Program (IIP) - Phase 1

An 8-40 GHz Wideband Instrument for Snow Measurements (WISM)

Institutional PI, \$4,000,000 total, \$94,947 to BSU

Collaborative proposal with Harris Corporation, NASA Goddard,
University of Washington, Nuvotronics LLC

08/2010-08/2012 NASA Terrestrial Hydrology Program

*Ground-based radar calibration / validation measurements for
NASA CLPX-II 2010, Grand Mesa, Colorado*

PI, \$35,468

08/2006-07/2009 NASA Energy and Water-cycle Sponsored (NEWS) Research
and Terrestrial Hydrology Program (THP)

*Investigation of Spatial and Temporal Variation in Snowpack Properties
Using Ground-Based High Resolution Microwave Radar Combined with
Detailed Snowpack Characterization*

PI, \$462,132 (\$114,232 moved to BSU when hired)

Co-I: Nick Rutter (Northumbria University, U.K), Rick Forster (University of Utah)

09/2005-08/2010 NSF Arctic Sciences Program

*Collaborative Research: Polarimetric Characteristics of Radio-wave Scattering from Water
Pathways within Glaciers: Laboratory Experiments and Computer Simulations*

PI, \$69,785

Co-PI: Kenichi Matsuoka (University of Washington)

Advising

Primary Advising

Name	Degree	Advisor Role	Program/University	Graduation
Joe Meyer	Research Scholar	Primary	06/23-present	
Achim Heilig	Post-Doc	Primary	Geophysics	01/11-12/12
Ibrahim Alabi	Post-Doc	Primary	Geophysics	03/25-present
Scott Havens	Ph.D.	Primary	Geophysics	Fall 2014
Andrew Hedrick	Ph.D.	Primary	Geophysics	Fall 2018
Tate Meehan	Ph.D.	Primary	Geophysics	Summer 2022
Chris McCastlin	Ph.D.	Primary	Geophysics	Spr 2024
Zach Keiskinen	Ph.D.	Co-Primary	Geophysics	Sum 2024
Ibrahim Alabi	Ph.D.	Co-Primary	Geophysics	Spr 2025
Naheem Adebisi	Ph.D.	Primary	Geophysics	Spr 2025
Hank Hetrick	Ph.D.	Primary	Geophysics	Spr 2016,ABD
Joel Gongora	Ph.D.	Primary	Geophysics	Spr 2021, ABD
Chago Rodriguez	Ph.D.	Primary	Geophysics	Spr 2018, ABD
Evi Ofekeze	Ph.D.	Co-Primary	Geophysics	exp. Fall 2025
Nicolas Allet	Ph.D.	Co-Primary	Université de Sherbrooke	exp. Spr 2026
Lindsay Stark	Ph.D.	Primary	Geophysics	exp. Spr 2027
Coleman Kane	Ph.D.	Primary	Geophysics	exp. Spr 2027
Julia Lober	M.S.	Primary	Geophysics	exp. Fall 2027
Tom Pe	M.S.	Primary	Electrical and Mechanical Engineering	Spring 2023
Mitch Creelman	M.S.	Primary	Geosciences	Fall 2022
Megan Mason	M.S.	Primary	Geophysics	Summer 2020
Peter Youngblood	M.S.	Primary	Hydrology	Summer 2020
Tate Meehan	M.S.	Co-primary	Geophysics	Spring 2018
Andrew Hedrick	M.S.	Primary	Geophysics	Fall 2013
Jesse Dean	M.S.	Primary	Geophysics	Fall 2016
Mark Robertson	M.S.	Primary	Geophysics	Fall 2016
Gabriel Trisca	M.S.	Primary	Computer Science	Spring 2014
Tom Pe	Undergrad	Primary	Applied Physics	Spring 2019
Peter Olsoy	Undergrad	Primary	Biology	Spring 2011
Jennifer Morris	Undergrad	Primary	Geosciences	Spring 2011
Jason Jennings	Undergrad	Primary	Geophysics	Spring 2012
Andrew Peterson	Undergrad	Primary	Geophysics	Spring 2012
Jake Robertson	Undergrad	Primary	Geosciences	Spring 2013
Rachel Parsons	Undergrad	Primary	Geosciences	Spring 2013
Mark Robertson	Undergrad	Primary	Geophysics	Fall 2012
Andrew Karlson	Undergrad	Primary	Geosciences	Spring 2016
Robert Florence	Undergrad	Primary	Geosciences	Spring 2015
Kris Burch	Undergrad	Primary	Electrical and Computer Engineering	Spring 2012
Lloyd Lowe II	Undergrad	Primary	Electrical and Computer Engineering	Spring 2012
Kolton Drake	Undergrad	Primary	Electrical and Computer Engineering	Spring 2012

Committee Member / Secondary Advising

Name	Degree	Advisor Role	Program/University	Graduation
Stanley Akor	Ph.D.	Committee	PhD Computing	exp Spring 2027
Adam Dickson	M.S.	Committee	Geophysics	exp Fall 2026
Isis Brangers	Ph.D.	Committee	Ku Leuven	Summer 2025
Eric Gagliano	Ph.D.	Committee	Civil Engineering, Univ. WA	exp Fall 2025
Mahboubah Boueshagh	Ph.D.	Committee	Lehigh Univ	exp Summer 2026
Rainey Aberle	Ph.D.	Committee	Geosciences/BSU	Spring 2025
Karina Zikan	Ph.D.	Committee	Geosciences/BSU	exp Spring 2027
Arash Modaresi Rad	Ph. D.	Committee	PhD Computing	Spring 2024
Ross Palomaki	Ph.D.	Committee	Geosci./Montana State Univ.	Summer 2023
Jack Tarricone	Ph.D.	Committee	Univ Nevada Reno	Summer 2023
Jukes Lui	Ph.D.	Committee	Geosciences/BSU	Spring 2024
Randall Bonnell	Ph.D.	Committee	Geosciences, Colorado State	Spring 2024
Brent Wilder	Ph.D.	Committee	Geophysics	Spring 2024
JB Madore	Ph.D.	Committee	Université de Sherbrooke, Département de Géomatique Appliquée	Spring 2024
Ahmad Hojatimalek	Ph.D.	Committee	Geosciences/BSU	Spring 2023
Chris Donahue	Ph.D.	Committee	Civil Eng./Montana State Univ.	Spring 2022
Isabelle Butler	M.S.	Committee	Civil Eng./BSU	Spring 2022
Jukes Lui	M.S.	Committee	Geosciences/BSU	Summer 2020
Colton Elkin	M.S.	Committee	Geosciences/BSU	Spring 2021
Rainey Aberle	M.S.	Committee	Geosciences/BSU	Spring 2021
Kate Bollen	M.S.	Committee	Geosciences/BSU	Spring 2021
Josh Morrel	M.S.	Committee	Geosciences/BSU	Spring 2021
Allison Vincent	M.S.	Committee	Geosciences/BSU	Summer 2020
Jewell Lund	M.S.	Committee	Geography/Univ Utah	Spring 2021
Will Rudisill	Ph.D.	Committee	Geosciences/BSU	Fall 2022
Maggi Kraft	Ph.D.	Committee	Geosciences/BSU	Spring 2022
Buchanan Kerswell	Ph.D.	Committee	Geosciences/BSU	Spring 2021
Miguel Aguayo	Ph.D.	Committee	Geosciences/BSU	Summer 2018
Katelyn Watson	Ph.D.	Committee	Geosciences/BSU	exp. Summer 2020
Jake Anderson	Ph.D.	Committee	Geophysics/BSU	Spring 2019
Kara Ferguson	M.S.	Committee	Geosciences/BSU	Spring 2018
Zakk Hess	M.S.	Committee	Geosciences/BSU	exp. Fall 2018
Zach Uhlmann	M.S.	Committee	Geosciences/BSU	Summer 2018
Megan Gallagher	M.S.	Committee	Geophysics/BSU	Summer 2018
William Rudisill	M.S.	Committee	Geosciences/BSU	Spring 2018
Lacie Rasley	M.S.	Committee	Geosciences/BSU	exp Spring 2018
Eric Lutz	Ph.D.	Committee	Geosci./Montana State Univ.	Spring 2009
James McCreight	Ph.D.	Committee	Civil Eng./Univ. Colorado	Fall 2011
Dylan Mikesell	Ph.D.	Committee	Geophysics/BSU	Fall 2011
Joel Brown	Ph.D.	Committee	Geophysics/BSU	Fall 2011
Eli Deeb	Ph.D.	Committee	Geography/Univ. Utah	Fall 2012
Thomas Blum	Ph.D.	Committee	Geophysics/BSU	Spring 2013
Patrick Kormos	Ph.D.	Committee	Geosciences/BSU	Fall 2013
Wade Tinkham	Ph.D.	Committee	Forest Ecol., Biogeosci./U of I	Spring 2013
Nicolas Kinar	Ph.D.	External Examiner	Hydrology, U of Saskatchewan	Fall 2013
Esther Babcock	Ph.D.	Committee	Geophysics/BSU	Spring 2015
Brian Terbush	M.S.	Committee	Geosciences/BSU	Spring 2015
Ravi Preeshageetha	M.S.	Committee	Computer Science/ BSU	Spring 2015
Josh Nichols	M.S.	Committee	Geophysics/BSU	Spring 2010
Emily Park	M.S.	Committee	Geophysics/BSU	Spring 2011
Alden Shallcross	M.S.	Committee	Geosciences/BSU	Fall 2011
Brian Anderson	M.S.	Committee	Geosciences/BSU	Spring 2011
David Eiriksson	M.S.	Committee	Geosciences/BSU	Spring 2012
Reggie Walters	M.S.	Committee	Geosciences/BSU	Fall 2013
Erik Boe	M.S.	Committee	Geosciences/BSU	Fall 2013

Teaching - Other Courses

01/2003-06/2003 T.A., *The Weather*, Program on Ocean and Atmospheric Sciences, Univ. CO

01/2000-06/2000 T.A., *Engineering Geology*, Dept. of Civil Engineering, Univ. CO

12/2002-12/2003 Instructor, *Avalanche Level I*, Alpine World Ascents

08/2001-06/2002 N.S.F. K-12 Outreach Fellow

Provided lectures/hands-on demonstrations to local 7th and 8th grade classes on glaciology and avalanche related topics; communicated via website with students during field season in W. Antarctica. Supervised high-school student on avalanche-related senior research project.

09/1999-03/2000 Volunteer High School Teacher, Old Crow, Yukon Territory

Developed and taught individualized curriculum for at-risk high school youth in this small isolated Arctic village of Gwitch'in First Nation people. Supervised construction on Youth Center, obtained funding for and organized week-long hunting and cultural activities field trip, created afterschool and evening basketball program. (see also Honors / Awards)

University Service

6/2019-present	Tenure and Promotion Committee, Geoscience Dept
8/2019-present	Tenure and Promotion Committee, College of Arts and Sciences
2011-2024	Five hiring committees, Geophysics Faculty
8/2018-2021	Steering committee for the Ph.D in Computing
1/2019-6/2019	Graduate Program Committee, Geophysics Rep.
8/2012-12/2015	Graduate Program Committee, Geophysics Rep.
9/2008-9/2011	Undergraduate Curriculum Committee, Geophysics Rep.
2012	Hiring committee, COAS Scientific Instrumentation Shop, Geosciences rep.
Fall 2010	Organized installation of INL software in Mini-Cave 3-D Visualization Laboratory
1/09-present	Established geodetic/survey grade GPS instrument pool for department
2010	Co-PI, BSU designated a NVIDIA CUDA HPC research center (one of 20 worldwide)
2011-present	Organize avalanche training and maintain pool of avalanche safety equipment for BSU

External Service

2021-present	NASA SnowEx Mission, Leadership team member
2023-2025	Guest editor, Natural Hazards and Earth Systems Sciences, ISSW 2023 Special Issue
2023	Co-organizer, International Snow Science Workshop, Bend, OR
2022-present	Science Alliance, Protect our Winters
2014-present	Winter Wildlands Alliance (WWA) Ambassador
2023- present	Community Snow Observations (CSO) Ambassador
2018-2021	NASA SnowEx Mission, Lead Project Scientist
2017	NASA SnowEx Mission, Leadership Team and co-lead of field campaign

2016	NASA Earth Venture Mission Panel Member (Earth observation satellites)
01/2015-present	Education board member, NASA SnowSchool
02/2015, 01/2016	Instructor, NASA/CUAHSI field school
12/2008-present	Session Chair, Fall AGU Meeting
Jan 6-8, 2012	Co-organizer, <i>SEG Cryosphere Geophysics Workshop: Understanding a Changing Climate with Subsurface Imaging</i> , Boise, ID
12/2008-01/2012	Cryosphere Section Representative, Fall AGU meeting <i>2 trips per year to Washington, D.C. to organize Cryosphere program (> 500 submissions annually) for largest geophysics meeting in U.S. (>15,000 submissions)</i>
2011	NASA Senior Review Panel Member (review 13 active Earth Science NASA Missions)
2010-2011	Lead Guest Editor: CRST special issue for the <i>2010 International Snow Science Workshop</i> (premier snow science meeting worldwide)
12/2010	Invited Keynote Speaker at the PRYN/APECS Career Development Workshop
03/2008,04/2009	Co-organized and taught Snow Characterization Workshop with Dr. Martin Schneebeli, Swiss Federal Institute for Snow and Avalanche Research (SLF) <i>3 day classroom and field-based workshop held in Silverton, Colorado in 2008 at the Center for Snow and Avalanche Studies, 10 invited graduate students and scientists from different U.S. universities. Invited to repeat workshop in 2009 in Manali, India, by the Indian Department of Defense Snow and Avalanche Study Establishment (SASE), more than 30 Indian Ph.D. scientists attended.</i>
01/2005-08/2011	Chair of Research Committee, American Avalanche Association <i>Administered four grants per year, only funding source specifically for snow avalanche research in U.S.</i>
01/2004-present	Journal Reviewer: <i>Cold Regions Science and Technology; Hydrologic Processes; Journal of Geophysical Research - Earth Surface; Journal of Geophysical Research - Atmospheres; Annals of Glaciology; Water Resources Research; Earth Surface Processes and Landforms; Journal of Glaciology; Transactions on Geoscience and Remote Sensing; Remote Sensing of Environment; Arctic, Antarctic, and Alpine Research; Geoscience and Remote Sensing Letters; Near Surface Geophysics; Northwest Science; Journal of Applied Geophysics</i>
04/2008-present	Proposal Reviewer: <i>NASA Terrestrial Hydrology Program (panel member); NSF Experimental Program to Stimulate Competitive Research (EPSCoR) program; NSF Arctic Sciences; NASA Cryospheric Sciences IceBridge (panel member); NSF Geophysics; NSF Hydrologic Sciences; American Avalanche Association</i>
2010	Organized and delivered <i>Snow Day</i> at the Discovery Center of Idaho
2011-present	Train volunteers and develop curriculum for Bogus Basin <i>Snow School</i>
06/2007-09/2009	Research Committee, Eastern Snow Conference
10/2009-09/2012	Research committee, American Institute for Avalanche Research and Education
09/2006	Session Chair, Fracture Mechanics, International Snow Science Workshop <i>also reviewed papers and helped plan this international conference; more than 800 total attendees</i>
2001-2005	Senior Research Project Mentor for 6 Boulder Valley HS students <i>Awards: 2002: Earth Sciences 1st Place, State and Nat. Finalist, Best in Show (Adam Maerz); 2001: Earth Sciences 3rd Place (John Powers)</i>

Teaching - BSU Courses Taught

Course Number/Name	Semester	Credits	# grad/u
GEOPH 367/567: Snow Science Field Methods	Spring 2025	2	9/1
GEOPH 522: Data Analysis and Geostatistics	Fall 2024	3	12/1
GEOPH 597: Cryosphere Writing Seminar	Fall 2023	1	7/0
GEOPH 567: Snow Science Field Methods	Spring 2022	2	10/0
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2021	3	20/3
GEOPH 566: Snow and Ice Physics	Spring 2021	3	6/0
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2020	3	24/5
GEOPH 597: CUAHSI Virtual University, Microwave Radar Remote Sensing	Fall 2020	3	4/0
GEOPH 567: Snow Science Field Methods	Spring 2020	2	1/0
GEOPH 597: CUAHSI Virtual University, Microwave Radar Remote Sensing	Fall 2019	3	10/0
GEOPH 597: Select Topics in Snow Science	Spring 2019	1	5/0
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2018	3	13/0
GEOPH 480: Research in Geophysics	Spring 2018	3	0/1
GEOPH 466/566: Snow and Ice Physics	Spring 2018	3	6/7
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2017	3	10/2
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2016	3	11/3
GEOS 486: Geosciences Capstone	Spring 2016	4	0/1
GEOPH 466/566: Snow and Ice Physics	Spring 2016	3	4/0
GEOPH 567: Snow Science Field Methods	Spring 2016	2	1/0
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2014	3	12/1
GEOPH 502: Processes and Properties of the Earth II	Spring 2014	4	4/0
GEOPH 467/567: Snow Science Field Methods	Spring 2014	2	1/1
GEOPH 480: Research in Geophysics	Spring 2014	3	0/1
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2013	3	16/0
GEOPH 502: Properties and Processes of the Earth II	Spring 2013	4	5/0
GEOPH 467/567: Snow Science Field Methods	Spring 2013	2	2/1
GEOPH 466/566: Snow and Ice Physics	Spring 2012	3	9/4
GEOPH 480: Research in Geophysics	Spring 2012	2	0/2
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2011	3	9/2
GEOPH 497/597: Snow Radar Remote Sensing	Fall 2011	1	7/2
GEOPH 480: Research in Geophysics	Fall 2011	2	0/2
GEOPH 497/597: Snow Science Field Methods	Spring 2011	2	5/7
GEOPH 497/597: Data Analysis and Geostatistics	Fall 2010	3	11/2
GEOPH 497/597: Snow and Ice Physics	Spring 2010	3	7/2
GEOPH 497/597, CE 497/597: Data Analysis / Geostat.	Fall 2009	3	18/1
GEOPH 305: Applied Geophysics, w/ John Bradford	Spring 2009	3	0/5
TOTAL		97	261/5

Computer Programming Skills

Windows, DOS, Macintosh, Unix, Linux operating systems

Matlab (Data Acq., Signal Proc., Statistics, PDE, Wavelet, Symbolic Math Toolboxes),

Developed GUI-driven analysis software for FMCW radar acquisition and processing, snowpit data display, and seismic signal detection of ice quakes.

C++, LaTeX, Python, Abaqus, IDL, GPSurvey, LabView, Awk, GMT, Pine, Emacs, Vi

Campbell Scientific data logger communications programs

S+, Mathematica, Surfer, IslandDraw, Excel, Access, Visual Basic

data acquisition, data analysis, signal processing, database management, finite element analysis

low power microcontrollers for ADC/DAC and data logging

Skills / Certifications

Avalanche Science - Rec Level 2+ certification (advanced, 2021)

Wilderness First Responder (2017-present)

CPR for the Professional rescuer (2017-present)

Technician Class HAM radio operator (call sign: KJ7LNN)

General Mobile Radio (GMRS) License (call sign: WRZT791)

UNAVCO InSAR training (2018)

First Responder (1998, EMT program, North Seattle C.C.)

Ground Penetrating Radar (monopulse and FMCW, 5 MHz - 18 GHz)

GPS + theodolite survey equipment, SnowMicroPenetrometer

Northwest Avalanche Institute - Level 1 certification (beginning)

Northwest Avalanche Institute - Level 2 certification (advanced)

American Institute for Avalanche Research and Education - Level 1 Instructor Training

Snow survival and crevasse rescue - U.S. Antarctic Program

experience with snowmobile operation and repair

experience with electronic, metal, and wood-working tools

geophysical instrument design and fabrication

Volunteer Activities

Special Olympics basketball coach (02/15-06/15)

ESL Teacher for refugees (08/09-07/10)

Mentor, Big Brothers / Big Sisters of La Plata County (07/2006-08/2008)

Volunteer high school teacher, Old Crow, Yukon Territory, Canada (1999-2000)

Helped install seismic station on the summit of Mt. Rainier (1999)

Logan School of Learning (2-3rd graders), Field Trip leader to Loveland Pass, CO (2003)

Co-taught 3-day English course to indigenous leaders in southern Chile (2003)

Guest lecturer, Nathan Hale High School Physics class (1999)

Notetaker for Disabled Student Services, University of Washington (1996)

Teen Feed (1995)

Affiliations

Member, Snow Radar Algorithm Team, NASA SCLP (2004-present)

International Glaciological Society (IGS)

American Geophysical Union (AGU, Cryosphere Representative 2008-2012)

Sigma Xi (Full member)

American Avalanche Association (Research Chair 2005-2011, Professional Member)

American Inst. for Avalanche Research and Education (AIARE) Certified Instructor (2003-04)

American Meteorological Society (Full Member)

Interests / Hobbies

Backcountry skiing / snowboarding, mountaineering, rock climbing, basketball, trail running, mountain biking

West African percussion, Flamenco percussion, yoga